Shared evidence | Crop breeding programme

Selective breeding: design a resilient crop

Name ______________________________ Date ________________

All numbers below are constructed sample data for a paper model. Yield and resistance have inherited components; neither is controlled by a single gene in this activity.

Round 1 rule: choose two parents. Both must have disease damage at most 2. At least one must have mean yield at least 45 g per plant. Use different family lines. Lower damage is better; 0 = no visible damage, 10 = severe damage.

Line / family	Mean grain mass (g/plant)	Disease damage (0-10)
A / F1	48	6
B / F2	40	2
C / F3	46	1
D / F1	52	8

Constructed sample data above: each line mean represents 10 plants in matched trial conditions.

Round 2 target: yield at least 45 g and damage at most 2. P-S are siblings from the selected cross; E is from a separate family. Constructed single-plant observations from one matched trial.

Candidate / family	Grain mass (g/plant)	Disease damage (0-10)
P / F4	45	1
Q / F4	48	6
R / F4	42	0
S / F4	50	2
E / F5 (unrelated)	45	1

W1. Parent choice and reasons. W2. Next offspring choice. W3. Its next partner and one check.

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Supported route | Record the decisions

Use the shared evidence. Say, write or dictate the same scientific explanation.

S1. Choose two first parents. Give one damage value for each and the qualifying yield value.

S2. Explain why A or D fails the programme rule even though it has high yield.

S3. Choose one offspring from P-S with high yield and low damage. Give both values.

S4. Choose sibling P or unrelated E as its next partner. Explain one reason and one limitation.

S5. Number the method in the correct order.

A. Compare offspring and select useful ones.

B. Select parents with desired inherited traits.

C. Repeat across generations.

D. Breed the selected parents.

S6. Finish: offspring still need testing because … .

Standard route | Two-generation breeding log

Use exact evidence from the shared page. Your log should make the decisions reproducible.

M1. State the programme goal. Choose round 1 parents using three numerical values and family-line evidence.

M2. Reject one alternative parent with a specific reason. Why is yield alone insufficient?

M3. Use the offspring evidence to select a next-generation candidate and partner. Explain the genetic-diversity decision.

M4. Explain how repeatedly selecting suitable parents can change later generations. Include inheritance and variation.

M5. Predict one possible outcome and describe how you would test it fairly before extending the programme.

Stretch route | Does the programme work?

Constructed sample data for a follow-up trial. Both lines used the same growing conditions; values are means of 10 plants. Selection line and comparison line began from the same mixed seed population.

Generation	Selected line mean yield (g/plant)	Comparison line mean yield (g/plant)
G0	34	34
G1	39	35
G2	43	35
G3	44	36

T1. Calculate the G0-G3 change for each line and the difference between those changes. What does the comparison add?

T2. Evaluate: “Every plant improved by 10 g because breeding guarantees success.” Use the table and its limitations.

T3. Propose two improvements to the evidence before recommending this crop: one about sampling/trial design and one about diversity or disease.

T4. Choose S with E or S with P from the shared data. Defend the decision while acknowledging one trade-off.

Exit and reflection | Explain the mechanism

Use short scientific statements. No family or medical examples are needed.

E1. Write the main four steps of selective breeding in order.

E2. Why test offspring before choosing the next parents?

E3. Name one risk of repeatedly breeding close relatives.

L1. What question did your audience ask? Which decision or explanation did you improve?

My next useful step is…

A. Read two trait columns together.

B. Explain selection across generations.

C. Explain genetic diversity as a risk issue.

D. Evaluate comparison data.